Snapshots help to store incremental backups regularly. We can store the snapshots in a repository, usually mounted on a local filesystem or a cloud-based service such as AWS S3, Microsoft Azure, or Google Cloud Platform. As figure 14.9 shows, administrators snapshot clusters on a regular basis to a storage medium and then restore them on demand.

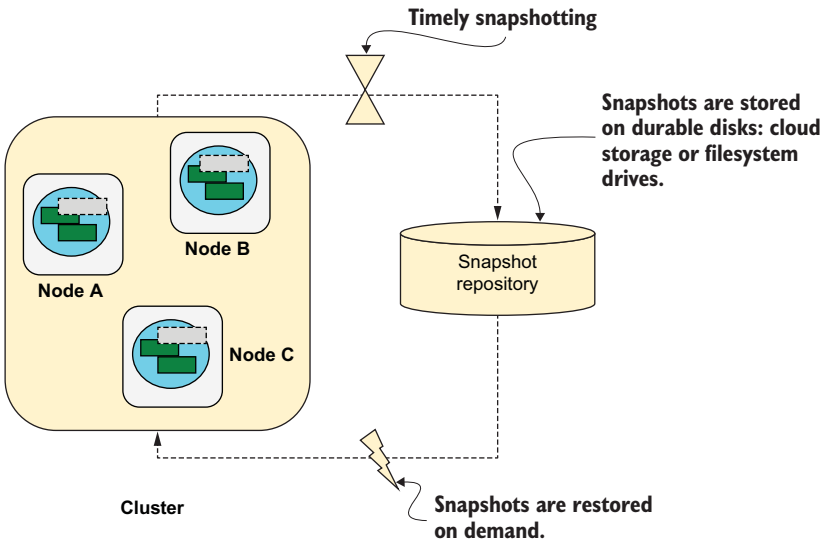


Figure 14.9 Mechanics of snapshot and restore on a cluster

A regular cluster snapshot is an administrative task that ideally should be automated with in-house scripts or handy tools. However, before we start backing up snapshots, we must ensure that our snapshot repository is a particular type and registered. This section discusses the mechanics to set up a repository and how to snapshot and restore data from the repository to a cluster.

14.4.1 Getting started

A few steps are required before we can use the snapshot and restore functionality. Broadly speaking, we need to perform the following three activities:

- *Register a snapshot repository*. Snapshots are stored on durable storage areas such as a filesystem, a Hadoop Distributed File System (HDFS), or cloud storage such as Amazon S3 buckets.
- *Snapshot the data*. Once we have a repository registered with the cluster, we can snapshot the data to back it up.
- *Restore from the store*. When we need to restore data, we simply pick the index, set of indexes, or full cluster that needs to be restored and start the restore operation from the previously registered snapshot repository.